

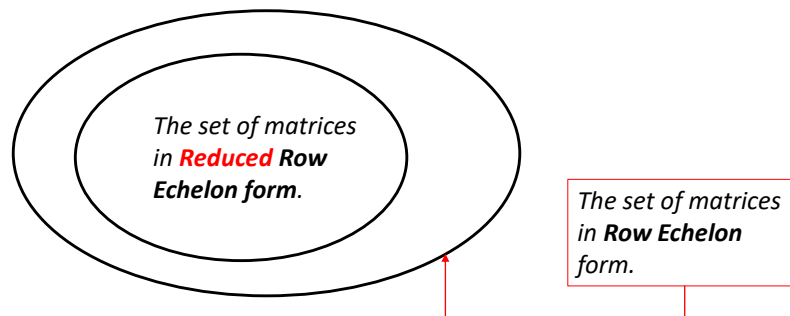
Week 3 Topic 1:

Reduced row echelon form
and Gauss-Jordan elimination.

Reduced Row Echelon form

[Video 3.1.1](#)

There is an even more simple form of a matrix, called reduced row echelon form. It is a matrix in row echelon form satisfying some extra rules.



Definition (Reduced Row Echelon form of a matrix.)

Let M be a matrix or an augmented matrix.

We say that M is in **Reduced Row Echelon form** if it is in row echelon form and in addition it satisfies the following two rules:

- A. The leading entry in every non-zero row is 1.
- B. Every column containing a leading entry of some row is zero in every position except the leading entry.

→ extra requirement from row echelon form

Rule A says:

“The leading entry in every non-zero row is 1.”

So every row is either all zero or looks like this:

$$\begin{bmatrix} 0 & \cdots & 0 & 1 & * & \cdots & * \end{bmatrix}$$

Rule B says:

“Every column containing a leading entry of some row is zero in every position except the leading entry.”

So if some row looks like this:

$$\begin{bmatrix} 0 & \cdots & 0 & 1 & * & \cdots & * \end{bmatrix}$$

then the other entries in the column containing that leading entry must all be zero:

$$\begin{bmatrix} 0 & \cdots & 0 & 0 & * & \cdots & * \\ 0 & \cdots & 0 & 1 & * & \cdots & * \\ 0 & \cdots & 0 & 0 & * & \cdots & * \\ 0 & \cdots & 0 & 0 & * & \cdots & * \end{bmatrix}$$

The rule insists these entries are zero. (We already know the entries below the 1 must be zero. Why?)

Video 3.1.2

Quick check of understanding:

Are any of the following matrices in REDUCED row echelon form?

(a) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ NO. This matrix is in row echelon form, but not in REDUCED row echelon form because the last column has an entry besides the leading term which is not zero.

(b) $\begin{bmatrix} 1 & 0 & 0 & 8 & 0 & 6 \\ 0 & 1 & 0 & 4 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 100 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ Yes. You can check it is in row echelon form, that all the leading entries are 1's, and that if a column has a leading entry then all its other entries are zero.

$\uparrow$ $\uparrow$
 not columns for leading entry

Quick check of understanding (cont.):

Are any of the following matrices in REDUCED row echelon form?

(c)
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$
 No. This matrix is not even in row echelon form because the leading entries don't move strictly to the right.

(d)
$$\begin{bmatrix} 1 & 0 & 10000 & 0 \\ 0 & 1 & 10 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$
 Yes. This matrix is in reduced row echelon form because it is in echelon form, the leading entries are all 1's, and moreover the columns containing a leading entry are otherwise all zeros.

↑
not a column
for leading
entry.

Video 3.1.3

Question:

Why is *Reduced Row Echelon form* even better than the normal row echelon form?

Answer:

The reason is that it makes the back-substitution step even simpler than it already is. When the system is reduced you don't have to do any calculations in the back-substitution step at all. You just rearrange the equations.

Let's see this by considering an example:

Example:

Consider the following augmented matrix and note that it is in reduced row echelon form:

$$\left[\begin{array}{cccccc|c} 1 & 0 & 23 & 0 & 1 & 0 & 1 \\ 0 & 1 & 7 & 0 & 62 & 0 & 10 \\ 0 & 0 & 0 & 1 & -5 & 0 & 100 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1000 \end{array} \right]$$

- 
- This column corresponds to variable x_6 .
 - The **1** at the bottom means we solve for this variable x_6 at the beginning of the back-substitution procedure.
 - Now because this matrix is in *reduced* row echelon form, every other entry in this column is zero, so we **never have to substitute the answer we get for x_6** into any other equation during the back-substitution procedure.

Example (continued):

In fact, because this matrix is in reduced row echelon form, the back-substitution procedure becomes trivial.

We can immediately write down the corresponding set of solutions immediately without any further calculation.

Example (continued):

First note that: As usual, variables corresponding to columns without leading entries become free parameters.

$$\begin{array}{cccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\
 \left[\begin{array}{cccccc|c}
 1 & 0 & 23 & 0 & 1 & 0 & 1 \\
 0 & 1 & 7 & 0 & 62 & 0 & 10 \\
 0 & 0 & 0 & 1 & -5 & 0 & 100 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1000
 \end{array} \right]
 \end{array}$$



We have two columns without leading entries, so we'll need two free parameters $s, t \in \mathbb{R}$. We set:

$$x_3 = s, \quad x_5 = t.$$

Example (continued):

Then: Just solve for variables corresponding to leading entries as is normal. Because the system is *reduced* we don't have to do any further substitutions.

$$\begin{array}{cccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\
 \left[\begin{array}{cccccc|c}
 1 & 0 & 23 & 0 & 1 & 0 & 1 \\
 0 & 1 & 7 & 0 & 62 & 0 & 10 \\
 0 & 0 & 0 & 1 & -5 & 0 & 100 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1000
 \end{array} \right]
 \end{array}$$



$x_3 = s \quad x_5 = t$

This equation says:

$$x_1 + 23x_3 + x_5 = 1.$$

So rearrange it to get:

$$x_1 = 1 - 23s - t.$$

This equation says:

$$x_2 + 7x_3 + 62x_5 = 10.$$

So rearrange it to get:

$$x_2 = 10 - 7s - 62t.$$

This equation says:

$$x_4 - 5x_5 = 100.$$

So rearrange it to get:

$$x_4 = 100 + 5t.$$

This equation says:

$$1x_6 = 1000.$$

Example (Conclusion):

If the linear system has the following augmented matrix in row reduced echelon form

$$\left[\begin{array}{cccccc|c} 1 & 0 & 23 & 0 & 1 & 0 & 1 \\ 0 & 1 & 7 & 0 & 62 & 0 & 10 \\ 0 & 0 & 0 & 1 & -5 & 0 & 100 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1000 \end{array} \right]$$

then we can immediately write down a parameterization for the set of solutions:

$$\begin{cases} x_1 = 1 - 23s - t \\ x_2 = 10 - 7s - 62t \\ x_3 = s \\ x_4 = 100 + 5t \\ x_5 = t \\ x_6 = 1000 \end{cases} .$$

Video 3.1.4

So if we can get the system to reduced row echelon form (**RREF** for short) then things are very simple...

... but how do we get to RREF then?

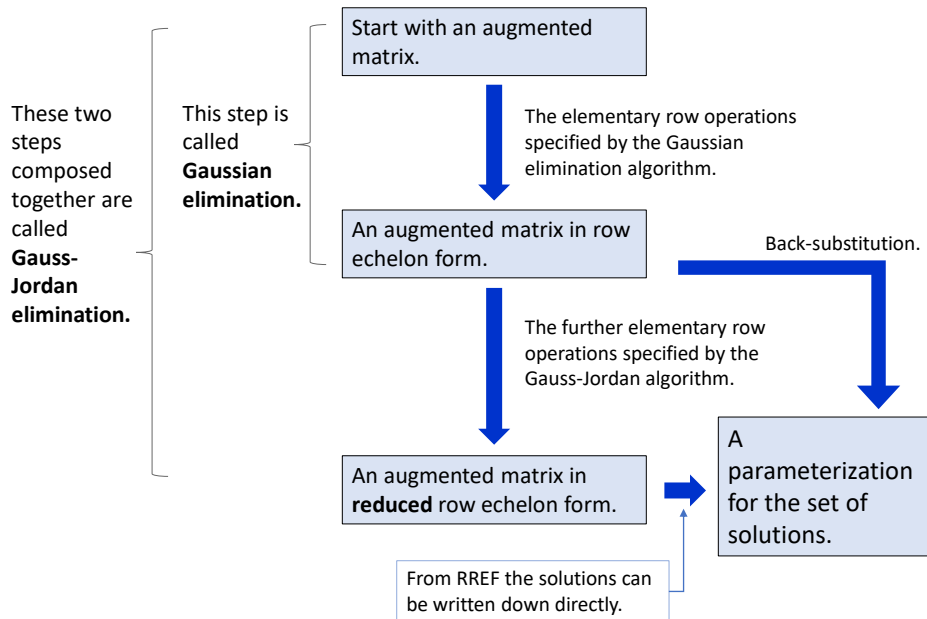
We'll see it's actually quite straightforward.

Gauss-Jordan Elimination

This is an algorithm which turns a matrix into RREF.

It is a **simple extension** of the Gaussian elimination algorithm using some steps to make the other entries in columns containing leading rows zero.

How the two methods relate to each other



The Algorithm called Gauss-Jordan Elimination.

The purpose of this algorithm is to turn any matrix or augmented matrix into, not just row echelon form, but *reduced* row echelon form.

We'll introduce this algorithm by following through the two examples below. (Note that these are the same two examples we used to introduce Gaussian elimination.)

$$\begin{bmatrix} 0 & 3 & \cdots & \cdots \\ 1 & -2 & \cdots & \cdots \\ 4 & 0 & \cdots & \cdots \end{bmatrix} \quad \begin{bmatrix} 0 & 3 & 1 & \cdots \\ 0 & 2 & -3 & \cdots \\ 0 & -1 & 6 & \cdots \end{bmatrix}$$

The Algorithm called Gauss-Jordan Elimination (cont.)

The first step is follow the algorithm for **Gaussian elimination** to its conclusion. After this you will have a matrix in row echelon form. (This is what Gaussian elimination does of course.)

$$\begin{bmatrix} 0 & 3 & \cdots & \cdots \\ 1 & -2 & \cdots & \cdots \\ 4 & 0 & \cdots & \cdots \end{bmatrix}$$



The sequence of elementary row operations specified by the Gaussian elimination algorithm.

$$\begin{bmatrix} 1 & -2 & \cdots & \cdots \\ 0 & 3 & \cdots & \cdots \\ 0 & 0 & \cdots & \cdots \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & \cdots \\ 0 & 2 & -3 & \cdots \\ 0 & -1 & 6 & \cdots \end{bmatrix}$$



The sequence of elementary row operations specified by the Gaussian elimination algorithm.

$$\begin{bmatrix} 0 & 3 & 1 & \cdots \\ 0 & 0 & -11/3 & \cdots \\ 0 & 0 & 0 & \cdots \end{bmatrix}$$

The Algorithm called Gauss-Jordan Elimination (cont.)

We follow that with an extra 2 steps.

Extra step #1: Multiply each non-zero row by a suitable constant to arrange the leading entry of the row to be 1.

No action is needed.

$$\begin{bmatrix} 1 & -2 & \cdots & \cdots \\ 0 & 3 & \cdots & \cdots \\ 0 & 0 & \cdots & \cdots \end{bmatrix}$$

Multiply the 2nd row by $1/3$ so that the entry marked by ● becomes 1.

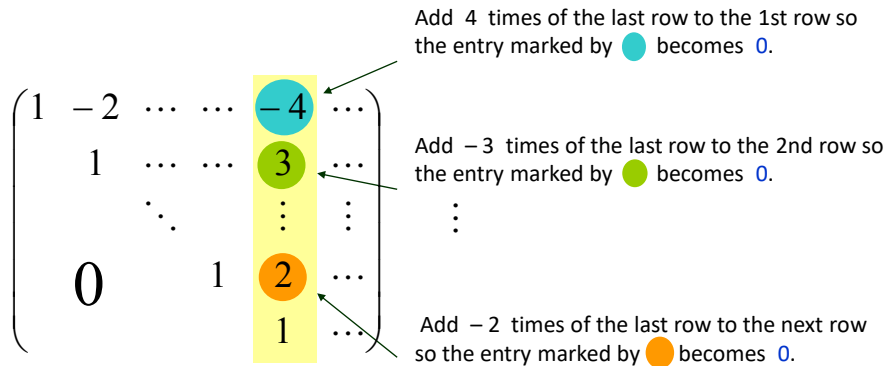
$$\begin{bmatrix} 0 & 3 & 1 & \cdots \\ 0 & 0 & -11/3 & \cdots \\ 0 & 0 & 0 & \cdots \end{bmatrix}$$

Multiply the 1st row by $1/3$ so that the entry marked by ● becomes 1.

Multiply the 2nd row by $-3/11$ so that the entry marked by ● becomes 1.

The Algorithm called **Gauss-Jordan Elimination** (cont.)

Extra step #2: Beginning with the last nonzero row, and working upward, add a suitable multiple of each row to the rows above to introduce zeros above the leading entries.



Continue up the matrix and do this for every row.
After these extra 2 steps the matrix will be in RREF.

Video 3.1.5

Example

We'll apply Gauss-Jordan elimination to find a sequence of elementary row operations which will turn the following matrix into row echelon form:

$$\begin{bmatrix} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix}.$$

Solution

The first step in the Gauss-Jordan algorithm is to apply the Gaussian elimination algorithm. You can check that this algorithm leads to the following sequence of elementary row operations resulting in the following row equivalent matrix in row echelon form:

$$\begin{bmatrix} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix}.$$



$$\begin{aligned} R_1 &\leftrightarrow R_2 \\ R_3 &\rightarrow R_3 + 2R_1 \\ R_3 &\rightarrow R_3 - \frac{3}{2}R_2 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{bmatrix}.$$

Solution (continued).

Then we do the two extra steps.

The first extra step is to make all the leading entries **1** by multiplying each row by a suitable scalar.

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{bmatrix}.$$



$$\begin{aligned} R_2 &\rightarrow \left(\frac{1}{2}\right) * R_2 \\ R_3 &\rightarrow \left(\frac{1}{6}\right) * R_3 \end{aligned}$$

$$\begin{bmatrix} \mathbf{1} & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & \mathbf{1} & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 & \mathbf{1} & -4 \end{bmatrix}.$$

Solution (continued).

The second extra step is to make every column containing a leading entry all zeros except for the leading entry.

In this case there are three columns we are going to have to focus on, corresponding to the three non-zero rows.

$$\begin{array}{c}
 \boxed{3} \quad \boxed{2} \quad \boxed{1} \\
 \downarrow \quad \downarrow \quad \downarrow \\
 \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}
 \end{array}$$

There are three columns containing leading entries we will have to "fix" in turn.

Solution (continued).

Fixing the column labelled 1 below:

$$\begin{array}{c}
 \boxed{3} \quad \boxed{2} \quad \boxed{1} \\
 \downarrow \quad \downarrow \quad \downarrow \\
 \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}
 \end{array}$$

$\downarrow$

$$\begin{array}{l}
 R_2 \rightarrow R_2 - R_3 \\
 R_1 \rightarrow R_1 - 3R_3
 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 0 & 3 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

Solution (continued).

Now fixing the column labelled 2 below:

$$\begin{array}{ccc}
 \boxed{3} & \boxed{2} & \boxed{1} \\
 \downarrow & \downarrow & \downarrow \\
 \begin{bmatrix} 1 & 2 & 4 & 5 & 0 & 3 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix} & & \\
 \downarrow R_1 \rightarrow R_1 - 4R_2 & & \\
 \begin{bmatrix} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix} & &
 \end{array}$$

Solution (continued).

Finally, fixing the column labelled 3 below:

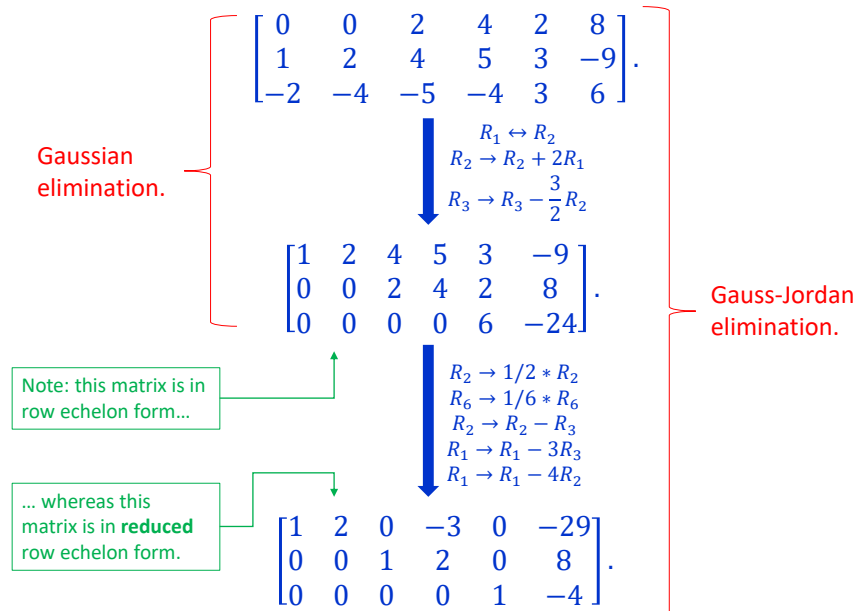
$$\begin{array}{ccc}
 \boxed{3} & \boxed{2} & \boxed{1} \\
 \downarrow & \downarrow & \downarrow \\
 \begin{bmatrix} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix} & &
 \end{array}$$

There is nothing to fix in this column, so we are done.

This matrix is row equivalent to the original matrix and is in reduced row echelon form.

(So the solutions of the corresponding system can be read off directly without further calculations.)

Summary



Let's do an exercise to check that the two methods do indeed lead to the same result.

Video 3.1.6

Exercise:

Consider the following system of linear equations:

$$\begin{cases} 2x_3 + 4x_4 + 2x_5 = 8 \\ x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 = -9 \\ -2x_1 - 4x_2 - 5x_3 - 4x_4 + 3x_5 = 6 \end{cases}$$

Note that the augmented matrix of this system is the same as the matrix we was just studying.

Compute the set of solutions of this system:

1. By Gaussian elimination followed by back-substitution.
2. By Gauss-Jordan elimination followed by writing down the solutions directly.

Solution

The first step is to write down the augmented matrix of the system:

$$\left[\begin{array}{ccccc|c} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right].$$

We computed the result of Gaussian elimination to be the following row echelon form:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{array} \right].$$

Solution (continued).

This is equivalent to the following system of linear equations:

$$\begin{cases} x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 = -9 \\ 2x_3 + 4x_4 + 2x_5 = 8 \\ 6x_5 = -24 \end{cases}$$

We solve this by back-substitution.

We start with the third equation: it gives us $x_5 = -4$.

Then we move to the 2nd equation. We solve for the variable corresponding to the leading entry x_3 after substituting $x_5 = -4$ into the equation and making x_4 a free parameter $x_4 = s$. We calculate: $x_3 = 8 - 2s$.

Solution (continued).

We are studying the following system of linear equations:

$$\begin{cases} x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 = -9 \\ 2x_3 + 4x_4 + 2x_5 = 8 \\ 6x_5 = -24 \end{cases}$$

Finally we move to the first equation. We will solve it for the variable corresponding to the leading entry which is x_1 .

First we substitute the solutions we have calculated so far, which are: $x_5 = -4$, $x_4 = s$, and $x_3 = 8 - 2s$. And also we assign a new free parameter to the remaining variable $x_2 = t$.

We calculate:

$$\begin{aligned} x_1 &= -9 - 2t - 4(8 - 2s) - 5s - 3 * (-4) \\ &= -29 + 3s - 2t. \end{aligned}$$

Solution (continued).

In summary, the result of solving the original system of linear equations using Gaussian elimination is the set of solutions:

$$\begin{cases} x_1 = -29 + 3s - 2t \\ x_2 = t \\ x_3 = 8 - 2s \\ x_4 = s \\ x_5 = -4 \end{cases}, \quad s, t \in \mathbb{R}.$$

Note that an alternative way of writing this set of solutions is:

$$\{(-29 + 3s - 2t, t, 8 - 2s, s, -4), s, t \in \mathbb{R}\} \subset \mathbb{R}^5.$$

↓
in vector notation

Solution (continued).

Now let's see what we get if we use Gauss-Jordan elimination instead of Gaussian elimination.

In this case the augmented matrix we end up with is the following one which in this case is in *reduced* row echelon form:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{array} \right].$$

Because this is in reduced row echelon form we can just write the set of solutions down directly, in the way we have learnt.

Solution (continued).

Recall and observe:

- The leading entries are marked with *red circles*.
- Variables corresponding to columns without leading entries contribute free parameters to the solution. (See the *green arrows*.)
- Then we use the equations corresponding to rows to solve for the variables corresponding to leading entries.

$$\left[\begin{array}{ccccc|c} \textcircled{1} & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & \textcircled{1} & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & \textcircled{1} & -4 \end{array} \right].$$



Solution (continued).

$$\begin{array}{ccccccccc}
 & x_1 & x_2 & x_3 & x_4 & x_5 & & & \\
 \left[\begin{array}{ccccc|c}
 \textcircled{1} & 2 & 0 & -3 & 0 & -29 \\
 0 & 0 & \textcircled{1} & 2 & 0 & 8 \\
 0 & 0 & 0 & 0 & \textcircled{1} & -4
 \end{array} \right]
 \end{array}$$

$\uparrow$ $x_2 = t$ $\uparrow$ $x_4 = s$

This equation says:

$$x_1 + 2x_2 - 3x_4 = -29.$$

So rearrange it to get:

$$x_1 = -29 - 2t + 3s.$$

This equation says:

$$x_3 + 2x_4 = 8.$$

So rearrange it to get:

$$x_3 = 8 - 2s.$$

This equation says:

$$x_5 = -4.$$

Conclusion.

We have found exactly the same set of solutions using Gauss-Jordan as we

found earlier using Gaussian elimination:

$$\begin{cases}
 x_1 = -29 + 3s - 2t \\
 x_2 = t \\
 x_3 = 8 - 2s \\
 x_4 = s \\
 x_5 = -4
 \end{cases}
 , \quad s, t \in \mathbb{R}.$$

Question:

Video 3.1.7

So which is better for solving linear systems: Gaussian Elimination or Gauss-Jordan elimination?

Discussion:

Well, it really depends on what you are trying to do.

If you are working with a small matrix there is no real difference in the time it takes.

But if you are putting a large system onto a computer then Gaussian elimination followed by Back substitution is actually the quickest algorithm.

On the other hand, if you are just calculating by hand, Gauss-Jordan may be better because reduced row-echelon form is simpler than plain row echelon form, so the back-substitution step has less chance of errors.

Later we'll meet other applications where one choice may be better than the other.

Video 3.1.8

Exercise

Given a **quadric surface** with equation

$$ax^2 + by^2 + cz^2 = d,$$

where a, b, c, d are real constants, that passes through the points

$$(1, 1, -1), (1, 3, 3) \text{ and } (-2, 0, 2),$$

use Gauss-Jordan elimination to find a formula for the quadric surface.

For $(1, 1, -1)$, For $(1, 3, 3)$ For $(-2, 0, 2)$

$$a+b+c-d=0 \quad a+9b+9c-d=0 \quad 4a+4c-d=0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 1 & 9 & 9 & 0 \\ 4 & 0 & 4 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 8 & 10 & 0 \\ 4 & 0 & 4 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 4R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 8 & 10 & 0 \\ 0 & -4 & 8 & 0 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -4 & 8 & 0 \\ 0 & 8 & 10 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 26 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow -\frac{1}{4}R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 26 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow \frac{1}{26}R_3} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + 2R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

If we choose $t=4$
 $4x^2 + 3y^2 - 3z^2 = 4$

Solⁿ: $a=t$ $b=\frac{3}{4}t$
 $c=\frac{3}{4}t$ $d=t$
 For every t ,
 $t x^2 + (\frac{3}{4}t) y^2 - (\frac{3}{4}t) z^2 = t$
 satisfies the conditions.

free parameter

Video 3.1.9

The reduced row echelon form of a matrix is **unique**

Remember the definition of **row equivalence**: Two matrices are said to be row equivalent when you can get from one to the other by a sequence of elementary row operations.

Question:

If we start with a matrix, we find a row equivalent matrix in row echelon form by doing Gaussian elimination.

But is there only one possible answer?

In other words, are there more than one matrices in row echelon form row equivalent to a given matrix?

The row echelon form is not a strict enough definition for there to only be only row echelon form given the starting matrix.

Exercise:

Give a few examples of different matrices which are

- row equivalent
- in row echelon form.

Solution:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + 5R_2} \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow -5R_2} \begin{bmatrix} 1 & 5 & 0 \\ 0 & -5 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

These matrices are

- all in row echelon form
- all row equivalent
- but different

But if we take it a step further and ask the same question for **reduced** row echelon form, something special happens:

Theorem (The RREF of a matrix is unique.)

Let M be a matrix or an augmented matrix.

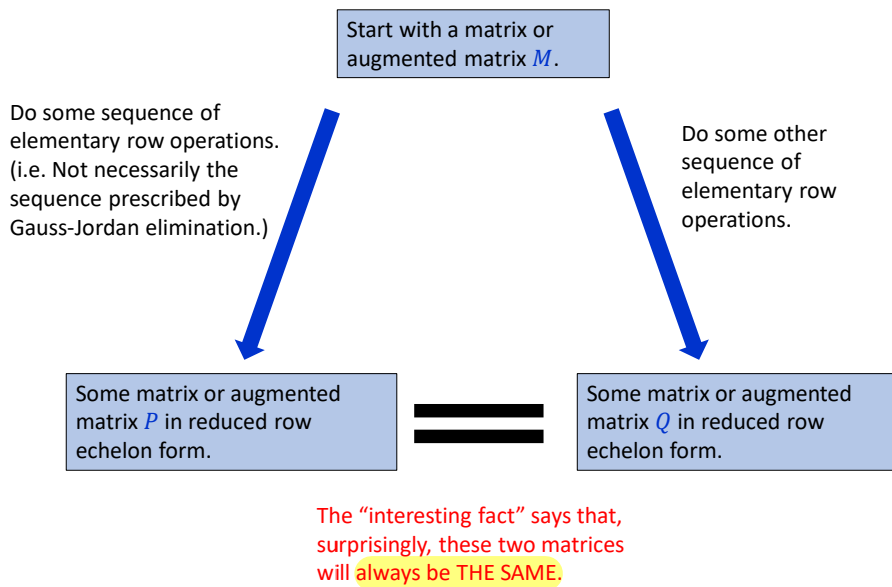
Let P and Q be matrices which are:

- in reduced row echelon form
- row equivalent to M .

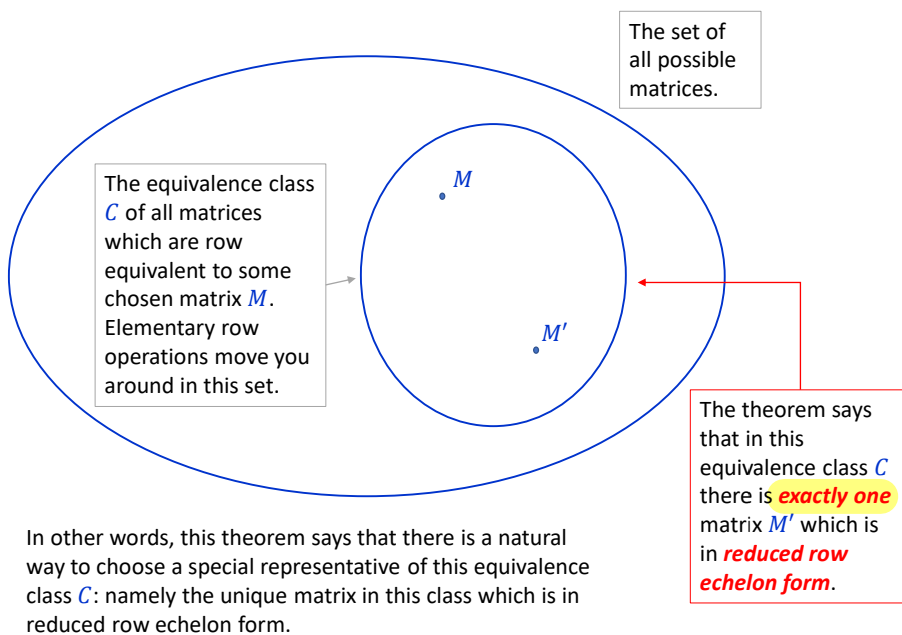
Then:

$$P = Q.$$

In other words...



Another way of thinking about this:



Week 3 Topic 2: “Counting” solutions.

Video 3.2.1

“Counting” solutions

Sometimes we just want to know “how many” solutions a system has and don’t actually need to go through a full computation of a parameterization of the set of solutions.

If that is all we want to know then, actually, a row echelon form of the augmented matrix of the system is all we need.

We will prove the following important theorem later on.

Definition (Three possibilities for the set of solutions.)

Consider a system of linear equations.

There are exactly three possibilities:

Possibility #1: A system has no solutions. (In other words it is an inconsistent system.)

Possibility #2: A system has exactly one solution. Unique solution

Possibility #3: A system has infinitely many solutions.

The problem we will discuss here: Consider a system of linear equations and let M denote a row echelon form of its augmented matrix. How can we decide which possibility holds for the system by looking at M ?

The problem we will discuss here: Consider a system of linear equations and let M denote a row echelon form of its augmented matrix. How can we decide which of the 3 possibilities holds for the system by looking at M ?

First question. Is the system consistent?

This is easy to check. Look at the last non-zero row of M .

The system is inconsistent if and only if it is of the following form (i.e. all zeros to the left of the partition, non-zero to the right of the partition).

$$\left[\begin{array}{ccc|c} * & \cdots & * & * \\ * & \cdots & * & * \\ 0 & \cdots & 0 & \neq 0 \\ 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{array} \right]$$

Check the last non-zero row in the row echelon form matrix M .

Why is this true?

Case 1: The last non-zero row of M looks like this:

$$\left[\begin{array}{ccc|c} * & \cdots & * & * \\ * & \cdots & * & * \\ 0 & \cdots & 0 & \neq 0 \\ 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{array} \right]$$

This system has **no solutions** because you will never be able to make the equation corresponding to that row true. The LHS is always zero, the RHS is always non-zero.

Case 2: The last non-zero row of M **doesn't** look like this:

$$\left[\begin{array}{ccc|c} * & \cdots & * & * \\ * & \cdots & * & * \\ 0 & \cdots & 0 & \neq 0 \\ 0 & \cdots & 0 & 0 \\ \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{array} \right]$$

This system has solutions because the matrix M is **in row echelon form** so back-substitution will definitely provide some solutions.

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 10 \\ 0 & 1 & 1 & 1 & 100 \end{array} \right]$$

→ back substitution will always get you solutions.

Next question. Assume the system is consistent. Does it have one or infinitely many solutions?

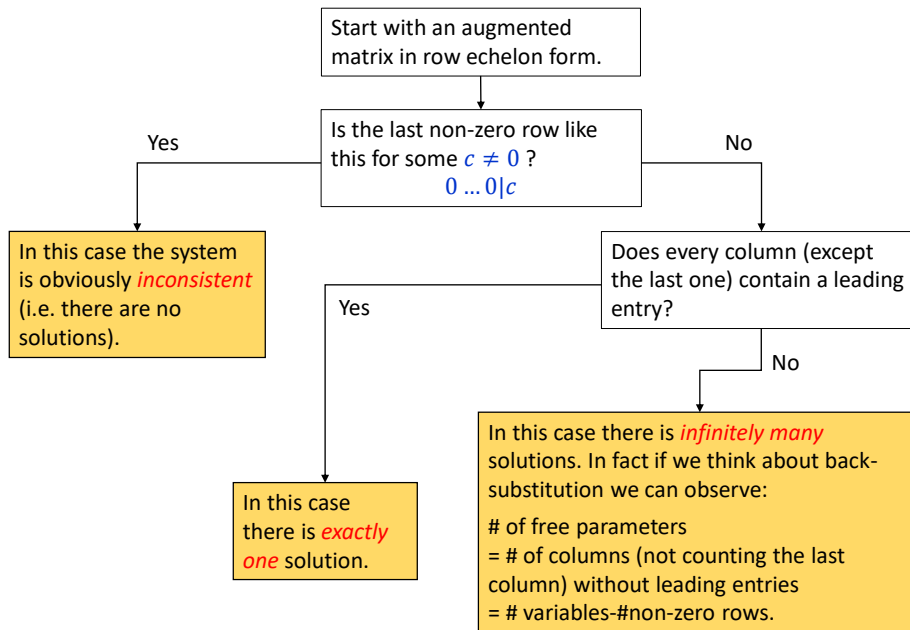
Again, this is easy to check **from a row echelon form** M of the augmented matrix.

You just have to understand whether back-substitution leads to free parameters in the parameterization.

The system will have infinitely many solutions if the following equivalent conditions are true:

- There is at least one column in M **without a leading entry**.
- There are more variables in the system than non-zero rows in M .

How many solutions? Flow chart summary



Let's do some examples.

Video 3.2.2

Example:

If a linear system has augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 5 \\ 0 & -1/2 & 4 & 0 \\ 0 & 0 & 6 & 1 \end{array} \right] \begin{array}{l} \text{every column has leading entry} \\ \leftarrow \text{not } 0 \dots 0 | 0 \end{array}$$

how many solutions does it have?

Solution

consistent.
No free parameters
 $\Rightarrow 1$ solⁿ

Example:

If a linear system has augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 5 \\ 0 & -1/2 & 4 & 0 & 0 \\ 0 & 0 & 6 & 2 & 1 \end{array} \right] \rightarrow \text{shows that it's consistent}$$

how many solutions does it have? ? no leading entry.

Solution

There is infinitely many solutions.

Example:

If a linear system has augmented matrix

$$\left[\begin{array}{ccccc|c} 1 & 1 & 2 & 1 & 0 & 5 \\ 0 & -0.5 & 4 & 0 & 1 & 15 \\ 0 & 0 & 6 & 2 & 2 & 10 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow \text{inconsistent form}$$

how many solutions does it have?

Solution

Zero solutions

Example:

If a linear system has augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{array} \right] \rightarrow \text{not in row echelon form.} \quad \text{must be in row echelon form to check}$$

how many solutions does it have?

Solution Do Gaussian Elimination to put into row echelon form

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & -1 \end{array} \right] \rightarrow \text{inconsistent!}$$

There are zero solutions.

Exercise:

If a linear system has augmented matrix

$$\left[\begin{array}{cccccc|ccc} 1 & 1 & 2 & 5 & 6 & 10 & 0 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 & 9 & -2 & 7 & 6 & 0 & 0 \\ 0 & 0 & 6 & 1 & 1 & 1 & -9 & 0 & 2 & 0 \\ 0 & 0 & 0 & -3 & 2 & -4 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 9 & 7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \text{consistent}$$

how many solutions does it have?

Infinitely many solutions.

no leading entry $\rightarrow$ free parameter

Solution

Is the following statement true or false?

Video 3.2.3

“A linear system with 3 variables and two equations can have exactly one solution.”

Solution

$$\begin{bmatrix} - & - & - & | & - \\ - & - & - & | & - \end{bmatrix}$$

possibilities

① Inconsistent

$$\begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 1 & 1 & 1 & | & 2 \end{bmatrix}$$

② Consistent

- Gaussian Elimination
 - 2 non-zero rows
 - ↳ infinite solutions
 - 1 non-zero row
 - ↳ infinite solutions

Solution

This statement is **false**.

To analyze this we need to understand what the *row echelon form* of the augmented matrix of the system will look like.

The original augmented matrix will have *3 columns to the left of the partition, and two non-zero rows*. Then we need to apply Gaussian elimination to it.

Gaussian elimination may reduce the number of non-zero rows, so the result of Gaussian elimination will have *3 columns to the left of the partition, and less than or equal to two non-zero rows*.

We conclude that no matter what happens during Gaussian elimination, we are left with a matrix in row echelon form with more variables than non-zero rows.

Such a matrix either (i) has a row with the “inconsistent” pattern, giving no solutions, or (ii) will have a free parameter, leading to infinitely many solutions.

Same Solution, briefly summarised:

There are two possibilities:

- No solutions (inconsistent system).

Example:
$$\begin{cases} x + y + z = 0 \\ x + y + z = 1 \end{cases}$$

- Infinitely many solutions when the system is consistent

$$\begin{aligned} \# \text{ variables} &> \# \text{ equations} \\ &= \# \text{ rows in augmented matrix} \\ &\geq \# \text{ nonzero rows in a row-echelon form} \end{aligned}$$

Thus columns without leading entries, thus free parameters, thus infinitely many solutions.

Let's consider the general possibilities

Video 3.2.4

What are all the possibilities in the following three cases:

1. The case (# variables) > (# equations).
2. The case (# variables) = (# equations).
3. The case (# variables) < (# equations).

Solution to Case 1.

This is basically the case we just discussed.

The only possibilities are that (i) The system is inconsistent or (ii) The system has infinitely many solutions.

Solution to Case 2 (# variables) = (# equations).

All three possibilities may happen:

- Inconsistent:
$$\begin{cases} x + y + z = 0 \\ x + 2y + 3z = 100 \\ x + y + z = 1 \end{cases}$$
- One solution:
$$\begin{cases} x = 0 \\ y = 100 \\ z = 1 \end{cases}$$
- Infinitely many solutions:
$$\begin{cases} x + y + z = 0 \\ x + 2y + 3z = 1 \\ 2x + 3y + 4z = 1 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 1 \\ 2 & 3 & 4 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{array} \right] \left. \begin{array}{l} \text{same, so next step to make} \\ \text{row echelon form will lead} \\ \text{to only two non-zero rows} \end{array} \right\}$$

Solution to Case 3 (# variables) < (# equations).

Again, all three possibilities may happen.

Why?

Because as you do Gaussian elimination, the number of non-zero rows may decrease. You may end up with a matrix in echelon form with all three possible “shapes”.

Let’s try and cook up a few examples.

Possibility #1: An inconsistent system.

$$\begin{cases} x + y = 0 \\ x + y = 1 \\ x + y = 2 \end{cases}$$

Solution to Case 3 (# variables) < (# equations).

Possibility #2: Exactly 1 solution.

$$\begin{cases} x + y = 0 \\ x + 2y = 1 \\ 2x + 3y = 1 \end{cases}$$

Gaussian elimination applied to the augmented matrix gives:

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 2 & 3 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

This system is in row echelon form, is consistent, and has a leading entry for every column. Thus there is exactly one solution.

Solution to Case 3 (# variables) < (# equations).

Possibility #3: Infinitely many solutions.

$$\begin{cases} x + y = 1 \\ 2x + 2y = 2 \\ 3x + 3y = 3 \end{cases}$$

Gaussian elimination applied to the augmented matrix gives:

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 3 & 3 & 3 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

This system is in row echelon form, is consistent, and has a column without a leading entry.

Thus there is a free parameter in the solutions.

Thus there are infinitely many solutions.

Video 3.2.5

“Counting” solutions when your system depends on some extra constants.

Sometimes our systems contain some extra constants.

For example: In the system below, the constant b is viewed differently to the system variables x, y, z . Different choices of b give different linear systems in x, y, z which are the variables we then solve for.

only 3 variables! $\rightarrow x, y$ and z

$$\begin{cases} x + 2y + z = 1 \\ y + 2z = 2 \\ (1 - b^2)z = b + 1 \end{cases}$$

Set $b = 0$.

$$\begin{cases} x + 2y + z = 1 \\ y + 2z = 2 \\ z = 1 \end{cases}$$

A system with exactly one solution.

Set $b = -1$.

$$\begin{cases} x + 2y + z = 1 \\ y + 2z = 2 \\ 0z = 0 \end{cases}$$

?
 no leading entry in this column

A system with infinitely many solutions.

What we want to do: understand how many solutions there are for all different choices of the constant b . The following is a typical problem:

Example

Determine the values of the constant b so that the system of linear equations

$$\begin{cases} x + 2y + z = 1 \\ y + 2z = 2 \\ (1 - b^2)z = b + 1 \end{cases}$$

has

- (a) no solution,
- (b) a unique solution, and
- (c) infinitely many solutions.

Discussion of the Example

The system we are studying has augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1-b^2 & b+1 \end{array} \right]$$

How to analyse?

We know that to understand the number of solutions of a system we need to put the augmented matrix into row echelon form and study the *pattern* of the echelon form (i.e. where the leading entries are, and whether the last non-zero row is consistent.)

This matrix *seems* to be in row echelon form, but note that there are a number of entries which depend on the constant b . The final pattern will depend on what is chosen for b . We have to go through every possibility.

Discussion of the Example (continued)

The system we are studying has augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1-b^2 & b+1 \end{array} \right]$$

This entry will

1. be 0 if $b = 1$ or $b = -1$.
2. be $\neq 0$ otherwise.

This entry will

1. be 0 if $b = -1$.
2. be $\neq 0$ otherwise.

Discussion of the Example (continued)

The system we are studying has augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1-b^2 & b+1 \end{array} \right]$$

So to cover all possibilities it will be enough to focus on three cases:

1. Case 1: $b = +1$.
2. Case 2: $b = -1$.
3. Case 3: Other b .

(How did we get this list exactly? Look at every entry in the matrix which depends on the constant b . Work out for which b the entry is zero. Every such value of b becomes a separate case. Also you need to consider the remaining cases left over, "every other b " where those matrix entries depending on b are all $\neq 0$. Also if some entry depending on b doesn't affect the final pattern of row echelon form you can ignore it.)

Discussion of the Example (continued)

The system we are studying has augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1-b^2 & b+1 \end{array} \right]$$

Case 1: $b = +1$.

In this case the matrix has the pattern

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{array} \right].$$

The bottom row is of the inconsistent form so there are **no solutions**.

Case 2: $b = -1$.

In this case the matrix has the pattern

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

This is in row echelon form, consistent, and there is a column without a leading entry. Thus the system has **infinitely many solutions**.

Case 2: $b \neq 1, -1$.

In this case the matrix has the pattern

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & \neq 0 & \neq 0 \end{array} \right].$$

This is in row echelon form, consistent, and every column has a leading entry. Thus this system has exactly **one solution**.

Discussion of the Example (summary.)

The system we are studying has augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1-b^2 & b+1 \end{array} \right]$$

We found that there are three cases:

1. Case 1: $b = +1$. In this case we found that there are no solutions, so the system is *inconsistent*.
2. Case 2: $b = -1$. In this case we found that there were *infinitely many solutions*.
3. Case 3: Other b . In this case we found the system has *exactly one solution*.



In the previous example the augmented matrix of the system was given to us already in row echelon form and we just had to study the patterns in the different cases.

However in general, the system **may not be in row echelon form yet**, and you'll need to do Gaussian elimination first. We'll do some examples of this now.

You can just leave the unknown constant as it is as you do Gaussian elimination, unless the constant becomes part of a leading entry. **In that case you need to be careful you don't divide by zero!**

This is an extra complication and we'll think about what to do in that case shortly.

Video 3.2.6

Example

Consider the following system of linear equations.

Note that there are three variables x, y, z and the system depends on three constants a, b, c .

$$\begin{cases} x & +2y & -3z & = & a \\ 2x & +6y & -11z & = & b \\ x & -2y & +7z & = & c \end{cases} \rightarrow \text{not in row echelon form}$$

What relationship between the constants a, b, c must hold so that the system is consistent (i.e. has at least one solution)?

Solution

The augmented matrix of the system is:

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 2 & 6 & -11 & b \\ 1 & -2 & 7 & c \end{array} \right].$$

To understand the number of solutions we'll do Gaussian elimination to put this matrix into row echelon form.

As always, Gaussian elimination starts with the first column:

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 2 & 6 & -11 & b \\ 1 & -2 & 7 & c \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b - 2a \\ 1 & -2 & 7 & c \end{array} \right]$$

Solution (cont.)

Continuing with Gaussian elimination:

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b-2a \\ 1 & -2 & 7 & c \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b-2a \\ 0 & -4 & 10 & c-a \end{array} \right]$$

Moving to the second column:

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b-2a \\ 0 & -4 & 10 & c-a \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b-2a \\ 0 & 0 & 0 & c+2b-5a \end{array} \right]$$

This matrix now appears to be in row echelon form.

Solution (continued.)

This augmented matrix now appears to be in row echelon form.

Finally we have to study the different patterns the row echelon form can have depending on the values of the constants.

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b-2a \\ 0 & 0 & 0 & c+2b-5a \end{array} \right]$$

Case 1: $c + 2b - 5a = 0$.

In this case the matrix has the pattern

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b-2a \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The system is consistent, in row echelon form, and there is a column without a leading entry. So there are *infinitely many solutions*.

Case 2: $c + 2b - 5a \neq 0$.

In this case the matrix has the pattern

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b-2a \\ 0 & 0 & 0 & \neq 0 \end{array} \right].$$

The bottom row is on the inconsistent form.

Thus this system has *no solutions*.

Solution (continued.)

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b - 2a \\ 0 & 0 & 0 & c + 2b - 5a \end{array} \right]$$

Note that these two entries can be 0 or $\neq 0$ depending on the constants, but these entries don't affect the final pattern of the row echelon form, so we can ignore them.

Solution (conclusion.)

We were studying the system:

$$\begin{cases} x + 2y - 3z = a \\ 2x + 6y - 11z = b \\ x - 2y + 7z = c \end{cases}$$

This system will have at least one solution if and only if

$$c + 2b - 5a = 0.$$



Video 3.2.7

Exercise

Determine the values of the constant b so that the system of linear equations

$$\begin{cases} x + 2y + z = 1 \\ 2x + by + 2z = 2 \\ 4x + 8y + b^2z = 2b \end{cases}$$

has

- (a) no solution,
- (b) a unique solution, and
- (c) infinitely many solutions.

Solution

The augmented matrix of the system has the form

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & b & 2 & 2 \\ 4 & 8 & b^2 & 2b \end{array} \right].$$

We'll do Gaussian elimination as usual:

$$\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & b-4 & 0 & 0 \\ 4 & 8 & b^2 & 2b \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 4R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & b-4 & 0 & 0 \\ 0 & 0 & b^2-4 & 2b-4 \end{array} \right]$$

Solution (continued)

This matrix appears to be in row echelon form.

But note that in this problem, there are *three* entries which will affect the final pattern of the row echelon form:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & b-4 & 0 & 0 \\ 0 & 0 & b^2-4 & 2b-4 \end{array} \right]$$

The final pattern will differ depending on whether these entries are 0 or $\neq 0$.

At least one of these 3 entries will be zero if b equals 4, 2, or -2 . So we treat these cases specially. For every other choice of b those three entries are non-zero, and that will be a separate case.

How many possibilities are there?

There are *four*. Namely $b = 4$, $b = 2$, $b = -2$, and all other b .

1. Case $b = 4$. The matrix becomes: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 12 & 4 \end{array} \right]$. We do a single row swap to put this in row echelon form, and see the matrix is consistent, and there is a column without a leading entry. Thus there are *infinitely many solutions* in this case.
2. Case $b = 2$. The matrix becomes: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$. Again, this case has *infinitely many solutions*.
3. Case $b = -2$. The matrix becomes: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & -8 \end{array} \right]$. In this case the bottom row is *inconsistent*, so there are *no solutions*.
4. Case $b \neq -2, 2, 4$. In this case: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & \neq 0 & 0 & 0 \\ 0 & 0 & \neq 0 & \neq 0 \end{array} \right]$. The matrix is in row echelon form, consistent, and every column has a leading entry. So there is *exactly one solution*.



Video 3.2.8

An extra complication: make sure you don't divide by zero.

During Gaussian elimination, sometimes you have to divide. If the number you are dividing by depends on an unknown parameter such as b you have to be careful that it is not actually zero! Division by zero *makes no sense*.

For example, imagine a system depending on some b . Imagine that somewhere during Gaussian elimination it looks like this:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & b & 2 \\ 0 & b+1 & 0 \end{bmatrix}$$

For example, imagine a system depending on some b . Imagine that somewhere during Gaussian elimination it looks like this:

$$\begin{bmatrix} 1 & * & * & * \\ 0 & b & * & * \\ 0 & b+1 & * & * \end{bmatrix}.$$

The Gaussian elimination algorithm tells you to next kill the $b+1$ term using the b term via this move:

$$\begin{bmatrix} 1 & * & * & * \\ 0 & b & * & * \\ 0 & b+1 & * & * \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{b+1}{b} R_2} \begin{bmatrix} 1 & * & * & * \\ 0 & b & * & * \\ 0 & 0 & * & * \end{bmatrix}$$

But hang on! In the case $b = 0$ we have just divided by zero!

What to do about this??

The solution is to split the computation into two cases at this point: (i) The case where that entry is zero (ii) The case where it is non-zero.

Example

Consider the following system of linear equations which depends on two constants a and b . Determine the values of the constants a and b so that the system of linear equations

$$\begin{cases} x + y + z = 5 \\ ay + z = 1 \\ by + z = 1 \end{cases}$$

has no solution, a unique solution, and infinitely many solutions.

Solution

The augmented matrix of the system is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a & 1 & 1 \\ 0 & b & 1 & 1 \end{array} \right].$$

An unthinking application of the Gaussian elimination algorithm would lead to the operation: $R_3 \rightarrow R_3 - \frac{b}{a}R_2$. This would be a division by zero if $a = 0$.

To deal with this possibility we split our deductions from this point on into 2 parts:

1. The case $a = 0$.
2. The case $a \neq 0$.

Solution (continued).

First case. The case that $a = 0$:

In this case the augmented matrix of the system becomes:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & b & 1 & 1 \end{array} \right].$$

We continue with the Gaussian elimination algorithm. The next step it tells us to do is to row swap $R_2 \leftrightarrow R_3$. We get:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & b & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right].$$

This appears to be in row echelon form. So we finish by considering the possible patterns that can arise for the row echelon form depending on the remaining parameter b .

Solution (continued).

First case. The case that $a = 0$ (continued).

In this case we get: $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & b & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right]$. For further analysis we focus on the entry which depends on the parameter b .

Subcase #1: $b = 0$.

In this case the augmented matrix becomes $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right]$ which

is row equivalent to $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$. This matrix is in row echelon

form, is consistent, and has a column without a leading entry. So in this case there are **infinitely many solutions**.

Solution (continued).

First case. The case that $a = 0$ (continued).

In this case we get: $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & b & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right]$. For further analysis we focus on the entry which depends on the parameter b .

Subcase #2: $b \neq 0$.

In this case the augmented matrix becomes $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & \neq 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right]$.

This matrix is in row echelon form, is consistent, and every column contains a leading entry. So in this case the system has a **unique solution**.

Solution (continued).

Second case. The case that $a \neq 0$:

In this case the augmented matrix of the system becomes:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a \neq 0 & 1 & 1 \\ 0 & b & 1 & 1 \end{array} \right].$$

We continue with the Gaussian elimination algorithm. Because we are assuming now that $a \neq 0$ the algorithm tells us to do the operation $R_3 \rightarrow R_3 - b/aR_2$. We get:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a \neq 0 & 1 & 1 \\ 0 & 0 & 1 - b/a & 1 - b/a \end{array} \right].$$

This appears to be in row echelon form. So we finish by considering the possible patterns that can arise for the row echelon form depending on the parameters b .

Solution (continued).

Second case. The case that $a \neq 0$:

We get:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a \neq 0 & 1 & 1 \\ 0 & 0 & 1 - b/a & 1 - b/a \end{array} \right].$$

The final analysis depends on the expression $1 - b/a$.

Subcase #1: This is the case that $1 - \frac{b}{a} = 0$. Because $a \neq 0$ this is equivalent to the condition $b = a$. In this subcase we are left with an augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a \neq 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

This matrix is in row echelon form, is consistent, and has a column without a leading entry. So the system has **infinitely many solutions** in this case.

Solution (continued).

Second case. The case that $a \neq 0$:

We get:

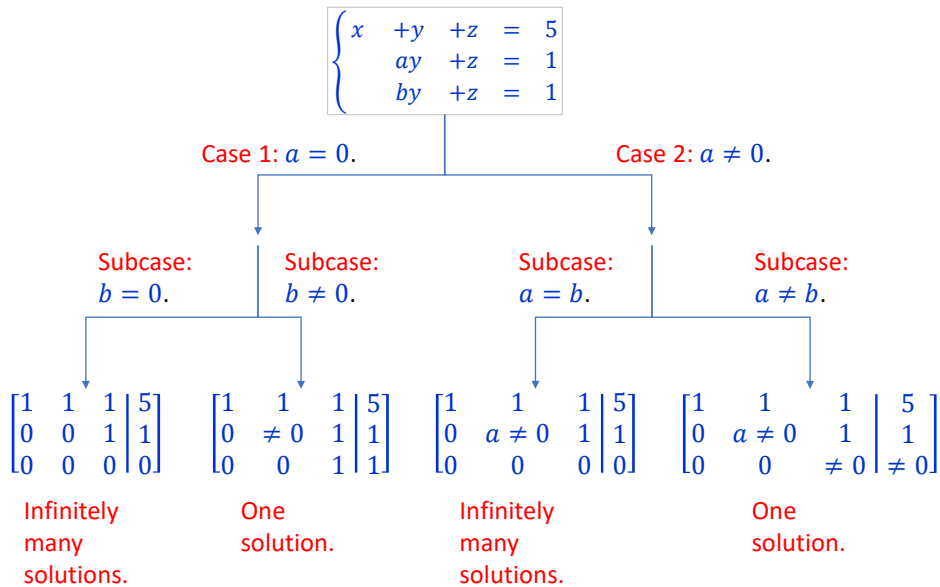
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a \neq 0 & 1 & 1 \\ 0 & 0 & 1 - b/a & 1 - b/a \end{array} \right].$$

The final analysis depends on the expression $1 - b/a$.

Subcase #2: This is the case $a \neq b$. In this subcase we are left with an augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a \neq 0 & 1 & 1 \\ 0 & 0 & \neq 0 & \neq 0 \end{array} \right].$$

This matrix is in row echelon form, is consistent, and every column has a leading entry. So the system has **exactly one solution** in this case.

Solution (Summary of the logic).**Solution (Summary).**

The following system

$$\begin{cases} x + y + z = 5 \\ ay + z = 1 \\ by + z = 1 \end{cases}$$

has solutions no matter what the values of a and b are.

If $a = b$ then there are infinitely many solutions.

If $a \neq b$ then there is exactly one solution.



This solution seems suspiciously simple after all that calculation... maybe there was an easier way to do this?

Remark:

Video 3.2.9

There are some tricks for understanding systems like this which can sometimes make the problem easier...

A Trick: Entries containing the constants make things more complicated. One trick is to do row swaps or column swaps to move those entries to less troublesome locations.

↓ changed the set of solutions

Example #1: A row swap.

Consider the following augmented matrix depending on two constants a and b :

$$\left[\begin{array}{ccc|c} a & 1 & 0 & a \\ 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \end{array} \right].$$

Gaussian elimination tells us to use the top left entry to kill the terms below it. This would be troublesome because we would have to split into two cases right away.

Instead it is easier to first do a row swap to move a simple 1 to the top left position:

$$\left[\begin{array}{ccc|c} a & 1 & 0 & a \\ 1 & 1 & 1 & 1 \\ 0 & 1 & a & b \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ a & 1 & 0 & a \\ 0 & 1 & a & b \end{array} \right]$$

Example #2: A column swap.

Please be much more careful when you do column swaps and other column operations!! The result is NOT a row equivalent matrix. The set of solutions changes.

Question:

Consider a linear system S with some ordering of the variables giving you an augmented matrix M .

Do a column swap $C_i \leftrightarrow C_j$ to some matrix M' .

Which system do you now have the augmented matrix of?

Answer:

It's the same linear system but with the ordering of the variables changed. The position of x_i has been swapped with the position of x_j .

Let's have another go at the system we solved earlier, and try and find an easier way.

Example #2: A column swap.

Consider the following system of linear equations which depends on two constants a and b . Determine the values of the constants a and b so that the system of linear equations

$$\begin{cases} x + y + z = 5 \\ ay + z = 1 \\ by + z = 1 \end{cases}$$

has no solution, a unique solution, and infinitely many solutions.

Example #2: (continued.)

If we assume that the variables are ordered in the obvious way $x \rightarrow y \rightarrow z$ then the augmented matrix for this system is as follows. Remember that the columns correspond to different variables.

$$\begin{array}{ccc|c} x & y & z & \\ \hline 1 & 1 & 1 & 5 \\ 0 & a & 1 & 1 \\ 0 & b & 1 & 1 \end{array}$$

We are going to do a column swap $C_2 \leftrightarrow C_3$ to make this easier to work with. We can do this but we have to remember that it corresponds to the same system but with the variables in a different order:

$$\begin{array}{ccc|c} x & y & z & \\ \hline 1 & 1 & 1 & 5 \\ 0 & a & 1 & 1 \\ 0 & b & 1 & 1 \end{array} \xrightarrow{C_2 \leftrightarrow C_3} \begin{array}{ccc|c} x & z & y & \\ \hline 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 1 & b & 1 \end{array}$$

Example #2: (continued.)

So the system we are studying has the following augmented matrix. We proceed with Gaussian elimination according to the algorithm.

$$\begin{array}{ccc|c} x & z & y & \\ \hline 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 1 & b & 1 \end{array} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{array}{ccc|c} x & z & y & \\ \hline 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 0 & b - a & 0 \end{array}.$$

To analyze this there are clearly two cases:

1. Case that $a = b$. In this case the augmented matrix becomes $\begin{bmatrix} 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ and we see there are infinitely many solutions.
2. Case that $a \neq b$. In this case the augmented matrix becomes $\begin{bmatrix} 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 0 & \neq 0 & 0 \end{bmatrix}$ and we see there is exactly one solution.



Note: this is exactly the same answer we obtained earlier by a direct method, but using this trick we got the answer much more quickly.

To analyze this there are clearly two cases:

1. Case that $a = b$. In this case the augmented matrix becomes $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$ and we see there are infinitely many solutions.
2. Case that $a \neq b$. In this case the augmented matrix becomes $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 0 & \neq 0 & 0 \end{array}\right]$ and we see there is exactly one solution.



A few comments about column operations.

- It didn't really matter that our variables were in a different order in the previous problem:

$$\begin{array}{c} x \quad y \quad z \\ \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & a & 1 & 1 \\ 0 & b & 1 & 1 \end{array}\right] \xrightarrow{C_2 \leftrightarrow C_3} \begin{array}{c} x \quad z \quad y \\ \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & a & 1 \\ 0 & 1 & b & 1 \end{array}\right] \end{array}$$

The reason is that we were only trying to “count” the number of solutions, not write them down explicitly.

If we had to write them down it would be easy to get the different parts of the solution in the wrong order when we write them down.

- Note that the other column operations like $C_i \rightarrow C_i + cC_j$ have a deeper impact on the system than this, and **can't be fixed easily by just “reordering” variables.**

not recommended to do.